What are we doing today?

2 Sessions + Pizza + Tea + Beer:

- Networking
- Session 1: Konstantin
- Session 2: Mirko
- Networking

[logo apache] | *logo-apache.png*

Welcome to the:

Apache Kafka Meetup

1000 Thanks! to our Sponsors:

- Klaustor Coworking - Halle (Saale)



- Confluent Inc.

[cflt logo] | *cflt-logo.png*

Kafka Meets Wayang in Halle (Saale), Germany

https://www.meetup.com/de-DE/leipzig-kafka/events/303683270/

How Apache Wayang Fosters Collaborative AI:

The Role of Data Federation for Assistant-Based Businesses

Dr. Mirko Kämpf <kamir@apache.org>

2024-11-12

Introduction

- This is a presentation for an example project.
- This project does something.
- We just haven't decided what that is yet.

Speaker Notes

Press the `s` key to access speaker notes.

You've unlocked the super secret notes area.

Abstract: For more than a decade we have heard that data is king, it was the next oil, and many other metaphors have been used. Besides having a bunch of buzzwords and data somewhere in the datacenter or the clouds, effectively leveraging distributed data resources becomes a cornerstone for competitive AI-driven businesses.

Especially when utilizing intelligent assistants is your goal - you probably are trapped in the data maze, where access and quality often hinder progress.

Apache Wayang might come to the rescue. The open source project enables collaborative machine learning and artificial intelligence through its advanced data federation capabilities across diverse processing platforms such as JVMs, Apache Spark, Apache Flink & Apache Kafka, and even traditional RDBMS. Apache Wayang allows a new approach to collaborative AI using decentralized feature stores.

Blank screen

You can press `b` or `.` to blank your current screen.

Hit it again to bring it back.

Overview

Press the `esc` key to get an overview of your slides.

Attributes

Press the down arrow key to see the next sub-slide.

Attributes Part 1

Built-in

asciidoctor-version

2.0.23

safe-mode-name

unsafe

docdir

/Users/kamir/GITHUB.active/incubator-training/content/Apache/Wayang/Apache-Wayang-Intro/src/main/asciidoc

docfile

/Users/kamir/GITHUB.active/incubator-training/content/Apache/Wayang/Apache-Wayang-Intro/src/main/asciidoc/index.adoc

imagesdir

../resources/images

Attributes Part 2

Custom

project-version

{project-version}

sourcedir

../java

Including documents from subdir

include::subdir/_b.adoc[]

content from *src/docs/asciidoc/subdir/_b.adoc*.

CAPTION-EX 1. include::_c.adoc[]

> content from *src/docs/asciidoc/subdir/_c.adoc*.

Images

[logo apache] | *logo-apache.png*

Videos

Play videos from Youtube, or locally

▶ https://www.youtube.com/watch?v=BKorP55Aqvg *(YouTube video)*

Lists

Asciidoctor allows a wide variety of list option.

For details see the documentation starting at: http://asciidoctor.org/docs/user-manual/#unordered-lists

Press the down arrow key to see the some examples.

Unordered List

- Protons
- Electrons
- Neutrons

Ordered List

1. Protons
2. Electrons
3. Neutrons

Nested List

Nest lists, by duplicating the list-char

- Protons
  - Electrons
- Neutrons

Tables

Header A	Header B
Cell A1	Cell B1
Cell A2	Cell B2
Cell A3	Cell B3

Source Code

Java code from project

```
public boolean contains(String haystack, String needle) {
    return haystack.contains(needle);
}
```

Two columns

- Blah
- Blah
- Blah
- Blah
- Blah
- Blah
- Blah
- Blah
- Blah
- Blah

Fontawesome Icons

- ✔
- 🔧
- 🍺

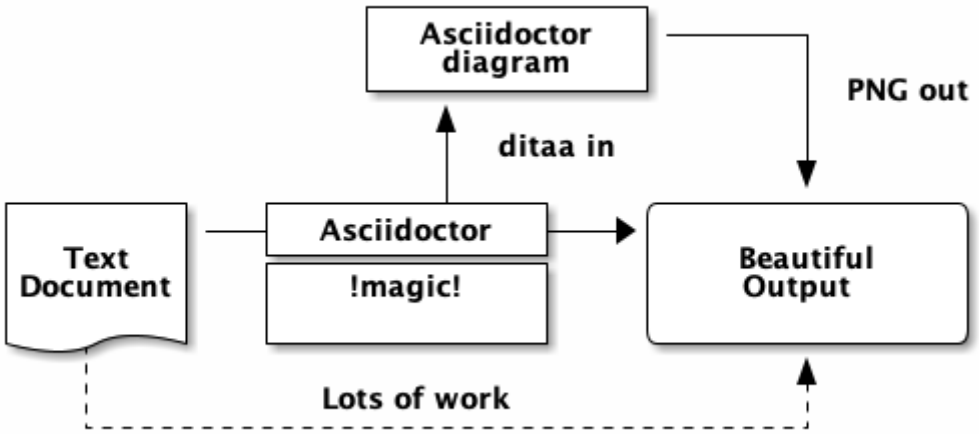


- Find icons here: https://fontawesome.com/v4.7.0/icons/
- Color icons by defining css styles in apache.css and referencing them with "role"

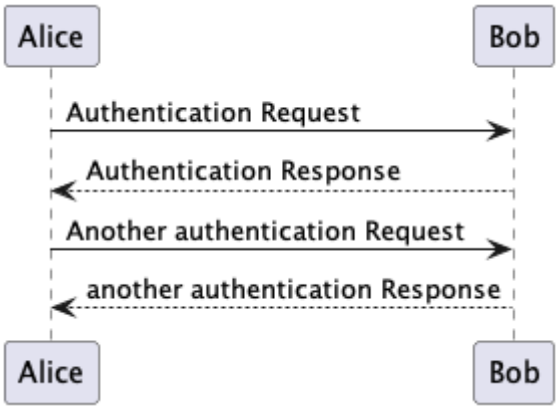
Diagrams

Press the down arrow key to see the different types of diagrams

Ditaa Diagram



PlantUML Diagram



GraphVIZ Diagram

```
Failed to generate image: Could not find the 'dot' executable in PATH; add it to the
PATH or specify its location using the 'graphvizdot' document attribute
digraph g {
    a -> b
    b -> c
    c -> d
    d -> a
}
```

Seqdiag Diagram

```
Failed to generate image: Could not find the 'seqdiag', 'seqdiag3' executable in PATH;
add it to the PATH or specify its location using the 'seqdiag' document attribute
{
    A -> B -> C;
         B -> D;
}
```

Blockdiag Diagram

```
Failed to generate image: Could not find the 'blockdiag', 'blockdiag3' executable in
PATH; add it to the PATH or specify its location using the 'blockdiag' document
attribute
{
   A -> B -> C -> D;
   A -> E -> F -> G;
}
```

Actdiag Diagram

```
Failed to generate image: Could not find the 'actdiag', 'actdiag3' executable in PATH;
add it to the PATH or specify its location using the 'actdiag' document attribute
{
  write -> convert -> image

  lane user {
     label = "User"
     write [label = "Writing reST"];
     image [label = "Get diagram IMAGE"];
  }
  lane actdiag {
     convert [label = "Convert reST to Image"];
  }
}
```

Nwdiag Diagram

```
Failed to generate image: Could not find the 'nwdiag', 'nwdiag3' executable in PATH;
add it to the PATH or specify its location using the 'nwdiag' document attribute
{
  network dmz {
      address = "210.x.x.x/24"
```

```
      web01 [address = "210.x.x.1"];
      web02 [address = "210.x.x.2"];
  }
  network internal {
      address = "172.x.x.x/24";

      web01 [address = "172.x.x.1"];
      web02 [address = "172.x.x.2"];
      db01;
      db02;
  }
}
```

Rackdiag Diagram

```
Failed to generate image: Could not find the 'rackdiag', 'rackdiag3' executable in
PATH; add it to the PATH or specify its location using the 'rackdiag' document
attribute
{
  // Change order of rack-number as ascending
  ascending;

  // define height of rack
  12U;

  // define description of rack
  description = "Tokyo/1234 East";

  // define rack units
  1: UPS [2U];   // define height of unit
  3: DB Server [5kg]  // define weight of unit
  4: Web Server [0.5A]  // define ampere of unit
  5: Web Server
  6: Web Server
  7: Load Balancer
  8: L3 Switch
}
```

Packetdiag Diagram

```
Failed to generate image: Could not find the 'packetdiag', 'packetdiag3' executable in
PATH; add it to the PATH or specify its location using the 'packetdiag' document
attribute
{
  colwidth = 32
  node_height = 72
```

```
  0-15: Source Port
  16-31: Destination Port
  32-63: Sequence Number
  64-95: Acknowledgment Number
  96-99: Data Offset
  100-105: Reserved
  106: URG [rotate = 270]
  107: ACK [rotate = 270]
  108: PSH [rotate = 270]
  109: RST [rotate = 270]
  110: SYN [rotate = 270]
  111: FIN [rotate = 270]
  112-127: Window
  128-143: Checksum
  144-159: Urgent Pointer
  160-191: (Options and Padding)
  192-223: data [colheight = 3]
}
```

Meme Diagram

```
Failed to generate image: Could not find the 'convert' executable in PATH; add it to
the PATH or specify its location using the 'convert' document attribute
```

Entity Relation Diagram

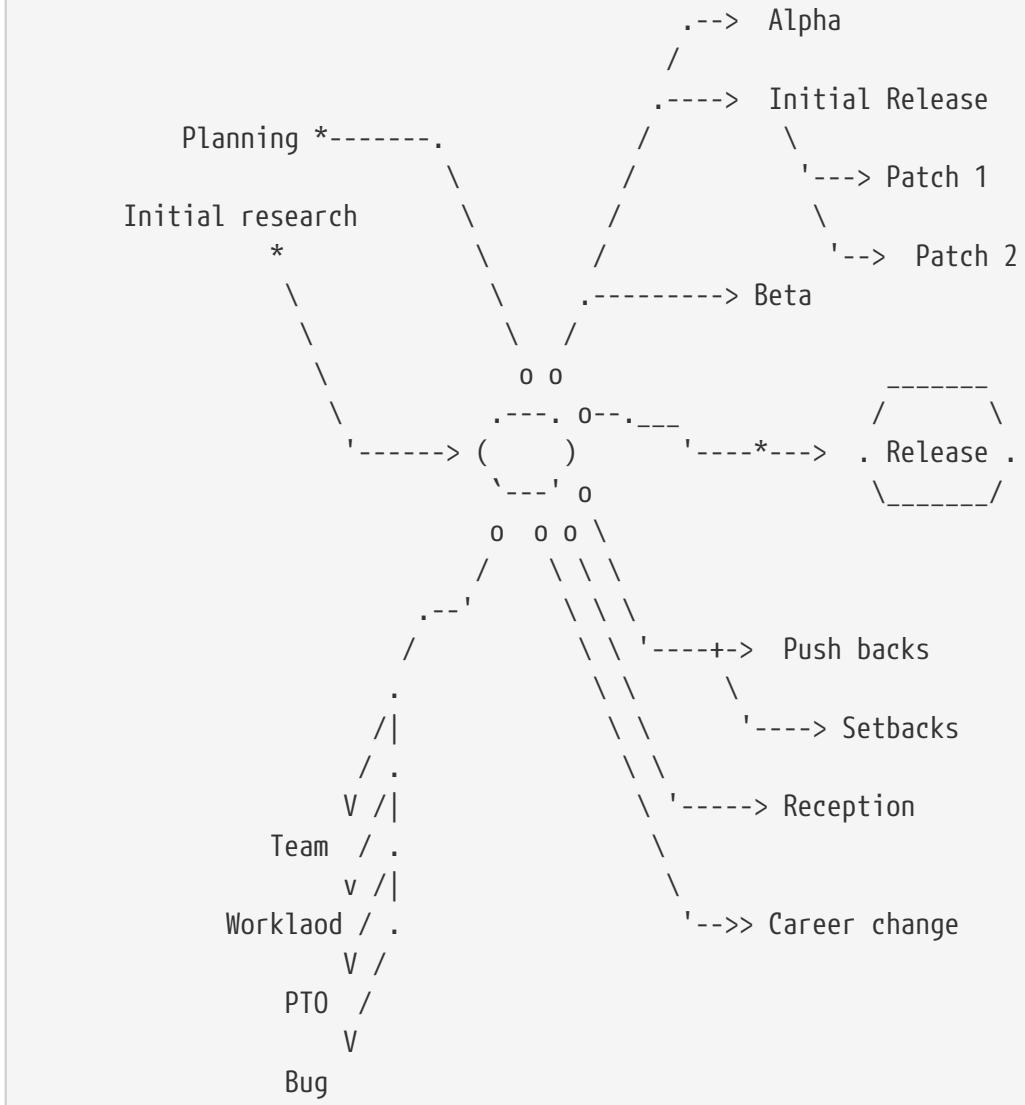
```
Failed to generate image: Could not find the 'erd', 'erd-go' executable in PATH; add
it to the PATH or specify its location using the 'erd' document attribute
[Person]
*name
height
weight
+birth_location_id

[Location]
*id
city
state
country

# Cardinality    Syntax
# 0 or 1         ?
# exactly 1      1
# 0 or more      *
# 1 or more      +
Person *--1 Location
```

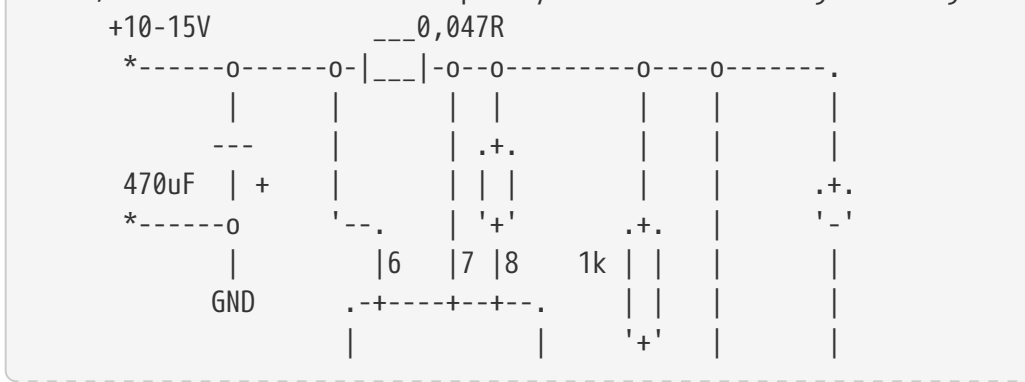
SVGBob: Mindmap

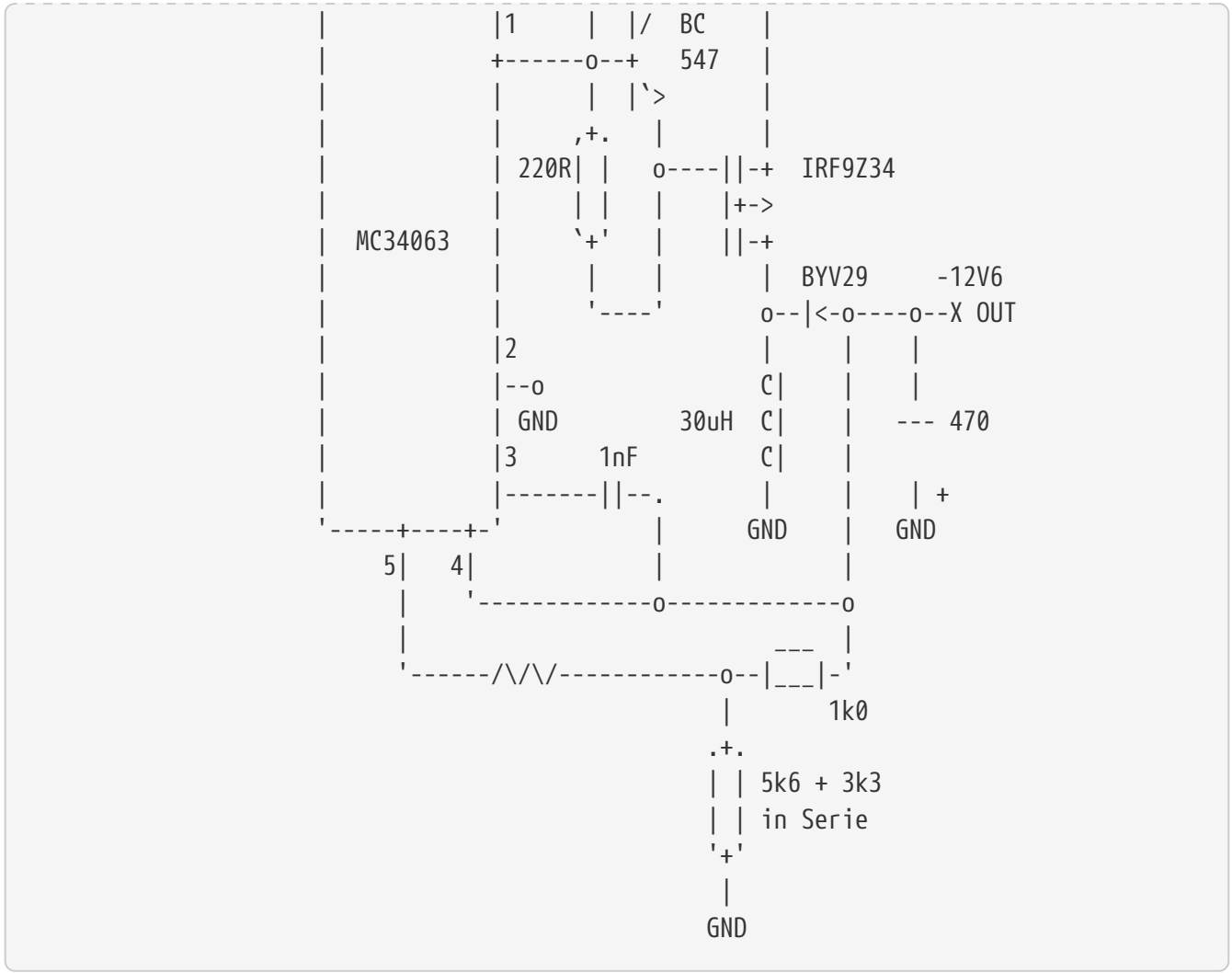
Failed to generate image: Could not find the 'svgbob', 'svgbob_cli' executable in PATH; add it to the PATH or specify its location using the 'svgbob' document attribute



SVGBob: Circuit Diagram

Failed to generate image: Could not find the 'svgbob', 'svgbob_cli' executable in PATH; add it to the PATH or specify its location using the 'svgbob' document attribute





Syntrax

```
Failed to generate image: Could not find the 'syntrax' executable in PATH; add it to
the PATH or specify its location using the 'syntrax' document attribute
indentstack(10,
  line(opt('-'), choice('0', line('1-9', loop(None, '0-9'))),
    opt('.', loop('0-9', None))),

  line(opt(choice('e', 'E'), choice(None, '+', '-'), loop('0-9', None)))
)
```

Embedded HTML/JavaScript

```
  <div id="iot-riskks1"></div>
  <script type="text/javascript">
c3.generate({
  bindto: '#iot-riskks1',
  size: { height: 500, width: 1000 },
  data: {
    x: 'x',
```

```
    columns: [
        ['x','Security','IT/OT Integration','Unclear ROI', 'Technical Expertise',
'Interoperability', 'Data Portability', 'Vendor Risk', 'Transition Risk',
'Legal/regulatory issues', 'Network constraints', 'Vendor lock-in'],
        ['%',43, 30, 28, 26, 26, 24, 22, 22, 22, 21, 18]
    ],
    type: 'bar'
  },
  axis: {
    rotated: true,
    x: {
      type: 'category',
      tick: {
        rotate: -90
      }
    }
  }
});
</script>
```